

Classful versus Classless IP Networks

IP Class	Range	Default Subnet Mask	Number of Networks	Number of Hosts per Network
A	1 – 126	255.0.0.0	126	16,777,214
B	128 – 191	255.255.0.0	16,384	65,534
C	192 – 223	255.255.255.0	2,097,152	254

Classful

Private Ranges:

Class A: 10.0.0.0 - 10.255.255.255

Class B: 172.16.0.0 - 172.31.255.255

Class C: 192.168.0.0 - 192.168.255.255

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Classful

Classless Inter-Domain Routing



Classless

IP Address	Subnet Mask	CIDR Notation
10.252.0.101	255.255.0.0	10.252.0.101/16

Private Address Space

The Internet Assigned Numbers Authority (IANA) has reserved the following three blocks of the IP address space for private internets:

10.0.0.0	-	10.255.255.255	(10/8 prefix)
172.16.0.0	-	172.31.255.255	(172.16/12 prefix)
192.168.0.0	-	192.168.255.255	(192.168/16 prefix)

We will refer to the first block as "24-bit block", the second as "20-bit block", and to the third as "16-bit" block. Note that (in pre-CIDR notation) the first block is nothing but a single class A network number, while the second block is a set of 16 contiguous class B network numbers, and third block is a set of 256 contiguous class C network numbers.

